constructed in this way; it does not need a time variable: it needs to tell us only how the things that we see in the world vary with respect to each other. That is to say, what the relations may be between these variables.[67]

The fundamental equations of quantum gravity are effectively formulated like this: they do not have a time variable, and they describe the world by indicating the possible relations between variable quantities.[68]

In 1967, an equation accounting for quantum gravity was written for the first time without any time variable. This equation was discovered by two American physicists—Bryce DeWitt and John Wheeler—and today it's known as the Wheeler–DeWitt equation.[69]

At first no one could understand the significance of an equation without a time variable, perhaps not even Wheeler and DeWitt themselves. (Wheeler: "Explain time? Not without explaining existence! Explain existence? Not without explaining time! To uncover the deep and hidden connection between time and existence . . . is a task for the future."[70]) The issue was discussed at great length; there were conferences, debates; rivers of ink flowed.[71] I think that the dust has now settled and things have become much clearer. There is nothing mysterious about the absence of time in the fundamental equation of quantum gravity. It is only the consequence of the fact that, at the fundamental level, no special variable exists.

The theory does not describe how things evolve *in time*. The theory describes how things change *one in respect to the others*,[72] how things happen in the world in relation to each other. That's all there is to it.

Bryce and John left us some years ago. I knew them both and had great admiration and respect for them. In my study at the university in Marseille, I have hanging on the wall a letter that John Wheeler wrote to me when he became aware of my first work on quantum gravity. Every so often I reread it with a mixture of pride and nostalgia. I would like to have asked him more, during the handful of meetings that we had together. The last time I went to see him in Princeton, we took a long walk together. He spoke to me with the soft voice of an old man: I could not make out much of what he said but did not dare to ask him too frequently to repeat what he was saying. Now he is no longer with us. I can't question him anymore, or tell him what I think. I can no longer tell him that it seems to me that his ideas are correct, and that they have guided me throughout a lifetime of research. I can no longer tell him I believe that he was the first to come close to the heart of the mystery of quantum gravity. Because he is no longer here—here and now. This is time for us. Memory and nostalgia. The pain of absence.